

Online Shopping System

SDLC - Planning Phase

This document describes the Planning Phase of the Software Development Life Cycle (SDLC) for a web-based Online Shopping System. The purpose of planning is to decide what the system will do, who will use it, what technology will be used, what resources are needed, and how the project will be completed.

1. Project Overview

The Online Shopping System is a web application that allows customers to browse products, search for products, manage a shopping cart, place orders, and use a simulated payment option. An administrator can manage products, users, stock, and orders.

2. Problem Statement

Traditional shopping requires customers to visit a physical store, search for products manually, and complete the purchase in person. The proposed system provides a simple online platform where customers can view products and place orders from a website. It also gives the administrator a central place to manage products and orders.

3. Project Objectives

- Create a modern and easy-to-use online shopping website.
- Allow customers to register, login, browse, search, and view products.
- Provide a shopping cart and checkout process.
- Allow customers to place and view orders.
- Provide a simulated payment method for academic purposes.
- Allow administrators to manage products, users, stock, and orders.
- Store application data in a MySQL database.
- Connect the HTML/CSS/JavaScript frontend with a Python backend.
- Build the system without frontend or backend frameworks.

4. Project Scope

In Scope	Out of Scope
User registration and login	Real banking/payment processing
Product browsing and search	Real financial transactions
Shopping cart	Delivery company integration
Checkout and simulated payment	Advanced AI recommendation system
Order creation and order history	Large-scale commercial deployment
Admin product management	Mobile native application
Admin order and user management	Third-party marketplace integration
MySQL database	Real card storage

5. Target Users

Customer: Searches and views products, manages a cart, checks out, and views orders.

Administrator: Manages products, users, stock, and order status.

6. Technology Plan

Layer	Technology	Purpose
Frontend	HTML5	Page structure and content
Frontend	CSS3	Layout, colors, responsive design, and visual styling
Frontend	Vanilla JavaScript	Validation, interactions, search, cart, and dynamic behavior
Backend	Python	Server-side logic and request handling
Database	MySQL	Store users, products, carts, orders, and related data

7. Major System Features

Module	Planned Features
Authentication	Register, login, logout, validation, password protection
Products	Product listing, details, categories, search, filtering, stock
Cart	Add, remove, quantity update, subtotal, total
Checkout	Customer details, shipping details, order summary, simulated payment
Orders	Create order, confirmation, order history, status
Profile	View and update customer information
Admin	Dashboard, product CRUD, users, orders, stock, status updates
Database	Users, admins, categories, products, cart, orders, order items

8. Project Development Plan

Stage	Work
Planning	Define problem, objectives, scope, users, resources, risks, and schedule.
Requirements Analysis	Identify functional and non-functional requirements and prepare requirement documentation.
Design	Prepare UI/UX, system architecture, database design, UML/DFD diagrams, and page layouts.
Implementation	Develop HTML, CSS, JavaScript, Python backend, and MySQL database.
Testing	Perform unit, integration, system, and user acceptance testing.
Deployment	Configure the server/database and make the application available for use.
Maintenance	Fix bugs, update products/data, back up the database, monitor the system, and improve features.

9. Basic Resource Plan

Human resources: student developer/team, project supervisor/teacher, and test users. Software resources: web browser, code editor, Python, MySQL, and a local development environment. Hardware resources: computer/laptop and internet connection when required. The project is designed to run locally during development.

10. Risk Planning

Risk	Possible Effect	Planned Solution
Database connection error	System cannot save/read data	Test connection early and keep database configuration separate.
Invalid user input	Incorrect data or crashes	Use frontend and backend validation.
Low stock	Orders may exceed available products	Check stock before adding to cart and placing orders.
Security weakness	Unauthorized access	Validate requests, protect admin functions, and hash passwords.
Broken frontend/backend components	Features do not work	Keep API names and data fields consistent and test each module.
Data loss	Orders/products may be lost	Create regular MySQL backups.

11. Planning Phase Deliverables

- Project overview and problem statement
- Project objectives and scope
- Target users
- Technology selection
- Major feature list
- Initial project schedule
- Resource plan
- Risk plan
- Initial SDLC roadmap

12. Planning Phase Conclusion

The planning phase establishes a clear direction for the Online Shopping System. The project will use HTML, CSS, and Vanilla JavaScript for the frontend, Python for the backend, and MySQL for data storage. The next SDLC step is Requirements Analysis, followed by Design, Implementation, Testing, Deployment, and Maintenance.